Polynomial Ring notes (From the Michael Penn Playlist)

Fuaad Amin

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1 The division algorithm for polynomials.

Let k be a field and $f(x), g(x) \in k[x]$. Then there are unique $q(x), r(x)$ such that $f(x) = g(x)q(x) + r(x)$ where

$$0 \leq \deg r(x) < \deg g(x)$$

For example in $\mathbb{N}$, we can conduct the division algorithm between 11 and 3 and we can choose $q = 3, r = 2$

$$11 = 3 \cdot 3 + 2 \implies 0 \leq 2 < 3$$

Doing the division algorithm with $4x^3 + 2x^2 + x + 3$ and $2x + 1$

$$\begin{array}{r} 2x^2 + \frac{1}{2} \\ 2x+1 \overline{)\,4x^3 + 2x^2 + x + 3} \\ \underline{-(4x^3 + 2x^2)} \\ x + 3 \\ \underline{-(x + \frac{1}{2})} \\ \frac{5}{2} \end{array}$$

Thus we see

$$4x^3 + 2x^2 + x + 3 = (2x + 1)\left(2x^2 + \frac{1}{2}\right) + \frac{5}{2}$$

Finally onto the proof.

Suppose $f(x), g(x)$ where $\deg(f) = n$ and $\deg(g) = m$ and suppose $m \leq n$
(otherwise $m > n \implies f(x) = 0 \cdot g(x) + f(x)$ and $\deg(f(x)) < \deg(g(x))$ and that's the trivial case done)
Proceed by induction ($n = m + k$). For the base case $k = 0$ we have $m = n$. Thus

$$f(x) = a_n x^n + ... + a_0$$

$$g(x) = b_n x^n + ... + b_0$$

We can rewrite $f(x)$ as

$$f(x) = f(x) + \overbrace{\frac{a_n}{b_n} g(x)}^{a_n x^n + ...} + \overbrace{\frac{a_n}{b_n} g(x)}^{a_n x^n + ...}$$

Hence

$$f(x) = \underbrace{\frac{a_n}{b_n}}_{q(x)} g(x) + \left[\underbrace{f(x) - \frac{a_n}{b_n} g(x)}_{r(x)}\right]$$

We know $\deg r(x) < n$ because both $f(x)$ and $\frac{a_n}{b_n} g(x)$ start with the term $a_n x^n$
For the induction hypothesis, suppose for all $p(x)$ where $\deg p(x) < n$ we can do the division algorithm. Consider

$$\hat{f}(x) = f(x) - \frac{a_n}{b_m} x^{n-m} g(x)$$

Expanding we see

$$(a_n x^n + ...) - \frac{a_n}{b_m} x^{n-m} (b_m x^m + ...)$$

Factoring out the x^n term and renaming the remaining terms

$$\left(a_n - \frac{a_n}{b_m} b_m\right) x^n + \hat{a}_{n-1} x^{n-1} + ... + \hat{a}_0$$

Thus $\hat{f}(x) = \hat{a}_{n-1}x_{n-1} + ... + a_0 \deg(\hat{f}) < n$. Next apply the induction hypothesis to $\hat{f}(x)$ and $g(x)$ we see that if we assume

$$\hat{f}(x) = g(x)\hat{q}(x) + \hat{r}(x)$$

Where $0 \leq \deg \hat{r}(x) < m$. Next we want to show that we can fit $\hat{f}(x)$ into a new equation involving $f(x)$ and we see

$$\begin{aligned} f(x) &= \hat{f}(x) + \frac{a_n}{b_m}x^{n-m}g(x) \\ &= g(x)\hat{q}(x) + \hat{r}(x) + \frac{a_n}{b_m}x^{n-m}g(x) \\ &= g(x)\left(\hat{q}(x) + \frac{a_n}{b_m}x^{n-m}\right) + \hat{r}(x) \\ &= g(x)q(x) + \hat{r}(x) \qquad (0 < \deg \hat{r}(x) < m) \end{aligned}$$

Finally we want to prove uniqueness. Suppose we have two ways to perform the division algorithm with $f(x)$ and $g(x)$. Then

$$f(x) = g(x)q_1(x) + r_1(x)$$
$$f(x) = g(x)q_2(x) + r_2(x)$$

where $0 \leq \deg r_i(x) < \deg g(x)$. Subtracting the equations

$$0 = g(x)\left(q_1(x) - q_2(x)\right) + \left(r_1(x) - r_2(x)\right)$$

$$r_2(x) - r_1(x) = g(x)\left(q_1(x) - q_2(x)\right)$$

Looking at the degree of LHS we see that $\deg(r_1(x) - r_2(x)) < \deg g(x)$. However on the RHS we see that $\deg \text{RHS} \geq \deg g(x)$ if $\deg(q_1 - q_2) > 0$. Hence in order to not contradict ourselves $q_1 - q_2 = 0$. and thus $r_1 - r_2 = 0$. Thus

$$q_1(x) = q_2(x) \qquad r_1(x) = r_2(x)$$

Thus we proved uniqueness of the quotient remainder

2 Writing the gcd of polynomials as a combination

Let k be a field and $d(x) \in k[x]$ is called the gcd of $f(x), g(x) \in k[x]$ if $d(x) \mid f(x)$ and $d(x) \mid g(x)$. If $\hat{d}(x) \mid f(x)$ and $\hat{d}(x) \mid g(x)$ then $\hat{d}(x) \mid d(x)$. If $\gcd(f(x), g(x)) = 1$ then we say $f(x)$ and $g(x)$ are relatively prime.

TheoremIf $d(x) = \gcd(f(x), g(x))$, then there exist $a(x), b(x) \in k[x]$ such that

$$a(x)f(x) + b(x)g(x) = d(x)$$

Consider the set

$$S = \{p(x)f(x) + q(x)g(x) \mid p(x), g(x) \in k[x]\}$$

Observe that there is a unique polynomial in S in which is monic and smallest degree because suppose that $u(x), v(x) \in S$ satisfy the property. Then clearly $u(x) - v(x) \in S$. But since they are monic and of equal degree, then their highest degree terms cancel out and we're left with $\deg(u-v) \leq n-1$. Thus we either have a contradiction and there is a smaller polynomial, or $u - v$ is the zero polynomial. And thus $u(x) = v(x)$. Take $d(x)$ to be the unique such monic and smallest degree polynomial. We want to claim that $d(x) \mid f(x)$. Using the division algorithm with $f(x)$ and $d(x)$

$$f(x) = d(x)q(x) + r(x)$$

Where $0 \leq \deg(r(x)x) < \deg(d(x))$. Rewriting we get

$$f(x) - d(x)q(x) = r(x)$$

Given that $d(x) \in S$ we can rewrite it as

$$r(x) = f(x) - (a(x)f(x) + b(x)g(x))\, q(x)$$

Combining $f(x)$ and $g(x)$ terms

$$r(x) = (1 - a(x)q(x))f(x) + (b(x)q(x))g(x) \in S$$

Notice that $r(x)$ is monic and $0 \leq \deg(r(x)x) < \deg(d(x))$. But we said that $d(x)$ has the least degree, meaning $r(x) = 0$ Meaning $f(x) = d(x)q(x)$ and thus $d(x) \mid f(x)$. We can perform the exact same argument to show that $d(x) \mid f(x)$. Suppose we had another divisor $\hat{d}(x) \in k[x]$ where $\hat{d}(x) \mid f(x)$ and $\hat{d}(x) \mid g(x)$. Hence we have $u(x), v(x) \in k[x]$ such that $f(x) = \hat{d}(x)u(x)$ and $g(x) = \hat{d}(x)v(x)$. Inserting them into the below equation

$$d(x) = a(x)f(x) + b(x)g(x)$$

we get

$$d(x) = a(x)u(x)\hat{d}(x) + b(x)v(x)\hat{d}(x)$$

$$d(x) = \hat{d}(x)(a(x)u(x) + b(x)v(x))$$

Thus $\hat{d}(x) \mid d(x)$ and thus

$$d(x) = \gcd(f(x), g(x))$$

3 Irreducible polynomials

We say a non-constant polynomial $f(x) \in k[x]$ is irreducible if it can't be written as $f(x) = g(x)h(x)$ where

$$\deg(g(x)), \deg(h(x)) < \deg(f(x))$$

Theorem $p(x) \in k[x]$ is irreducible $\iff$ $k[x]/\langle p(x)\rangle$ is a field where $\langle p(x)\rangle$ denotes the ideal generated by $p(x)$.

Suppose $p(x) \in k[x]$ is irreducible. Consider an ideal $I \subseteq [x]$ where

$$\langle p(x)\rangle \subsetneq I \subseteq k[x]$$

Let $f(x) \in I \setminus \langle p(x)\rangle$. Since $p(x)$ is irreducible and $f(x)$ is not a multiple of $p(x)$, then $\gcd(f(x), p(x)) = 1$. Thus there exists $a(x), b(x) \in k[x]$ such that

$$f(x)a(x) + p(x)b(x) = 1$$

Note that $a(x)f(x) \in I$ and $b(x)p(x)$. Thus $1 \in I$. Since a unit inside an ideal implies the entire ideal is the entire ring, then $I = k[x]$ meaning $\langle p(x)\rangle$ is maximal. Modding out a ring by a maximal ideal gives us a field. Thus we showed that

$$p(x) \in k[x] \text{ is irreducible } \implies k[x]/\langle p\rangle \text{ is a field}$$

For the converse, Suppose $k[x]/\langle p\rangle$ meaning $\langle p\rangle$ is a maximal ideal. Suppose that $p(x) = f(x)g(x)$. Thus

$$p(x) \in \langle f(x)\rangle$$

And thus

$$\langle p(x)\rangle \subseteq \langle f(x)\rangle \subseteq k[x]$$

However by $\langle p(x)\rangle$ being maximal, we have

$$\langle p(x)\rangle = \langle f(x)\rangle \quad \text{or} \quad \langle f(x)\rangle = k[x]$$

Suppose $\langle p(x)\rangle = \langle f(x)\rangle$ then thus $f(x) = p(x)h(x)$. But since $f(x)$ and $p(x)$ are multiples of each other, then $f(x)$ and $p(x)$ have equal degree. Thus $p(x)$ is irreducible. If $\langle f(x)\rangle = k[x]$ then $f(x)$ is a unit in $k[x]$. Thus $f(x)$ is a constant α. Hence $p(x) = \alpha g(x)$. The degree of $g(x)$ fails to be less than the degree of $p(x)$. Hence $p(x)$ is irreducible. Hence we proved that

$$p(x) \in k[x] \text{ is irreducible } \iff k[x]/\langle p\rangle \text{ is a field}$$

We can apply this and show that $\mathbb{C}$ is a field. Define $\varphi : \mathbb{R}[x] \to \mathbb{C}$ by $f(x) \mapsto f(i)$. We clearly see that it's an onto homomorphism and that $\ker\varphi = \langle x^2 + 1\rangle$. Since $x^2 + 1$ is irreducible in $\mathbb{R}$. By the first isomorphism theorem for rings, we see that

$$\mathbb{C} \cong \mathbb{R}[x]/\langle x^2 + 1\rangle$$

Since we know that $\mathbb{R}[x]/\langle x^2 + 1\rangle$ is a field, then $\mathbb{C}$ is a field.

We can show that $\mathbb{Q}(\sqrt{2})$ is a field. Define $\varphi : \mathbb{Q}[x] \to \mathbb{Q}(\sqrt{2})$. Where $f(x) \mapsto f(\sqrt{2})$. We see that $\ker\varphi = \langle x^2 - 2\rangle$. Which is irreducible in $\mathbb{Q}[x]$. Thus

$$\mathbb{Q}[x]/\langle x^2 + 2\rangle \cong \mathbb{Q}(\sqrt{2})$$

Finally consider $\mathbb{Z}_2[x]/\langle x^2 + x + 1\rangle$. We see that $x^2 + x + 1$ is irreducible (since 0 or 1 are not roots). Thus we constructed a field of order $4{=}2^2$.

4 Eisenstein's criterion

Lemma: Let $p(x) \in \mathbb{Q}[x]$. Then $p(x)$ can be written in terms of a polynomial with integer coefficients

$$p(x) = \frac{r}{s}\left(a_0 + a_1 x + ... + a_n x^n\right)$$

Where $\gcd(r, s) = 1$ and $\gcd(\{a_i\}) = 1$.

Let $p(x) = \frac{b_0}{c_0} + \frac{b_1}{c_1}x + ... + \frac{b_n}{c_n}x^n$ for $b_i, c_i \in \mathbb{Z}$. Factoring out the denominators we get

$$p(x) = \frac{1}{c_0 c_1 ... c_n}\left(d_0 + d_1 x + ... + d_n x^n\right), d_k = b_k c_0 c_1 ... c_{k-1} c_{k+1} ... c_n$$

Let $d = \gcd(d_0, ...d_n)$. Then we can write $d_0 = da_0$, $d_1 = da_1$ and so on. Thus

$$p(x) = \frac{d}{c_0 c_1 ... c_n}\left(a_0 + a_1 x + ... + a_n x^n\right)$$

Reducing the faction on the left to lowest terms, say r and s. Then

$$p(x) = \frac{r}{s}\left(a_0 + a_1 x + ... + a_n x^n\right)$$

Gauss' Lemma Let $p(x) \in \mathbb{Z}[x]$ be monic that factors $p(x) = \alpha(x)\beta(x) \in \mathbb{Q}[x]$ where $\deg \alpha(x), \deg \beta(x) < \deg P(x)$. Then there exists $a(x), b(x) \in \mathbb{Z}[x]$ such that $a(x), b(x)$ are monic, $\deg a(x) = \deg \alpha(x)$, $\deg b(x = \deg \beta(x)$ and

$$p(x) = a(x)b(x)$$

Suppose $p(x) = \alpha(x)\beta(x)$ where $\alpha(x), \beta(x) \in \mathbb{Q}[x]$. By the first lemma we can rewrite each polynomial as

$$\alpha(x) = \frac{c_1}{d_1}(a_0 + ... + a_m x^m) = \frac{c_1}{d_1}\alpha_1(x)$$

$$\beta(x) = \frac{c_2}{d_2}(b_0 + ... + b_n x^n) = \frac{c_2}{d_2}\beta_2(x)$$

Letting $\frac{c_1 c_2}{d_1 d_2} = \frac{c}{d}$

$$p(x) = \alpha(x)\beta(x) = \frac{c}{d}\alpha_1(x)\beta_1(x)$$

Rewriting one more time we get

$$dp(x) = c\alpha_1(x)\beta_1(x)$$

Case 1: $d = 1$
$\alpha_1(x), \beta_1(x) \in \mathbb{Z}[x]$. We also know that $Coeff_{x^{m+n}} p(x) = 1 = ca_m b_n$ If $c = 1$ then we let $a(x) = \alpha_1(x)$ and $b(x) = \beta_1(x)$ or we let $a(x) = -\alpha_1(x)$ and $b(x) = -\beta_1(x)$. If $c = -1$ then $a(x) = -\alpha_1(x)$ and $b(x) = \beta_1(x)$ or we let $a(x) = \alpha_1(x)$ and $b(x) = -\beta_1(x)$

Case 2: $d \neq 1$
Then there is a prime $p \mid d$ and since $\gcd(c, d) = 1$ then we can choose p such that $p \nmid c$. We also want to choose a_l such that $p \nmid a_l$ and b_k where $p \nmid b_k$. Set

$$\left.\begin{array}{l} \hat{\alpha} \equiv \alpha_1(x) \mid \mathbb{Z}_p[x] \\ \hat{\beta} \equiv \beta_1(x) \mid \mathbb{Z}_p[x] \end{array}\right\} \alpha(x), \beta(x) \neq 0 \text{ in } \mathbb{Z}_p[x]$$

Hence

$$\hat{\alpha}(x)\hat{\beta}(x) \equiv \alpha_1(x)\beta_1(x) = \frac{d}{c}p(x)$$

However since $p \mid d$ and $p \nmid c$ we get that the equation above is $\equiv 0 \mod \mathbb{Z}_p[x]$. However $\mathbb{Z}_p[x]$ is an integral domain. Meaning no zero divisors. Thus d must be equal to 1. Thus we achieved our desired factors.

Eisenstein's criterion: Let p be a prime and $f(x) = a_0 + ... + a_n x^n \in \mathbb{Z}[x]$ if $p \mid a^i$ for all $i \in \{0, 1, ..., n-1\}$, but $p \nmid a_n$ and $p^2 \nmid a_0$. Then $f(x)$ is irreducible of $\mathbb{Q}[x]$.

It would suffice to show that $f(x)$ does not factor over $\mathbb{Z}[x]$

Proof: Suppose $f(x)$ can be factored into

$$f(x) = (b_0 + ... + b_r x^r)(c_0 + ... + c_s x^s)$$

We notice that $p^2 \nmid a_0$ and $p \mid a_0$ with $a_0 = b_0 c_0 \implies p \nmid b_0$ or $p \nmid c_0$. Suppose WLOG that $p \nmid b_0$ and $p \mid c_0$. We also see that $p \nmid a_n$ where $a_n = b_r c_s \implies p \nmid b_r$ and $p \nmid c_s$. Let m be minimal such that $p \nmid c_m$. Consider a_m which we can write as

$$a_m = \underbrace{b_0 c_m}_{p\nmid} + \underbrace{b_1 c_{m-1} + ... + b_m c_0}_{\text{divisible by } p}$$

Thus a_m is not divisible by p thus $a_m = a_n \implies m = n$. Meaning we did not factor into two polynomials of lesser degree with integer coefficients. Meaning, $f(x)$ is irreducible in $\mathbb{Z}[x]$.

Examples: (I will include a few more)

$$3x^6 + 25x^5 - 20x^2 + 15x - 10$$

We see that it's irreducible with $p = 5$

$$5x^3 + 14x^2 - 7x + 7$$

We see that it's irreducible with $p = 7$

$$x^3 - 9x^2 + 15x - 6$$

We see that it's irreducible with $p = 3$

$$x^4 + 4x^3 + 6x^2 + 2x + 1$$

While not immediately following Eisenstein's criteria, we note that if it were reducible then we can shift the polynomial and preserve reducibility.

$$f(x) = \alpha(x)\beta(x) \implies f(y - c) = \alpha(y - c)\beta(y - c)$$

Since the above polynomial looks very close to $(x + 1)^4$ we shift by $x \mapsto y - 1$. Getting us

$$(y-1)^4 + 4(y-1)^3 + 6(y-1)^2 + 2(y-1) + 1$$

after some expanding and number pushing

$$y^4 - 2y + 2$$

And we can clearly see that it's irreducible with $p = 2$

5 Writing a polynomial gcd as a combination

We want to find $a(x), b(x)$ such that

$$a(x)f(x) + b(x)g(x)$$

Where

$$f(x) = x^4 - 2x^3 - 3x - 2$$
$$g(x) = x^3 + 4x^2 + 4x + 1$$

Over $\mathbb{Q}[x], \mathbb{Z}_2[x], \mathbb{Z}_{11}[x]$
He then proceeds to do a lot of long division and the euclidean algorithm and then we get

$$f(x) = (x-4)g(x) + (10x^2 + 12x + 2)$$

Then

$$g(x) = \left(\frac{1}{10}x + \frac{7}{25}\right)(10x^2 + 12x + 2) + \frac{11}{25}(x+1)$$

Again

$$10x^2 + 12x + 2 = (10x+2)(x+1)$$

Thus we found that $x+1$ is the GCD Thus we can use the equations for $f(x)$ and $g(x)$ above to show that

$$x + 1 = \frac{25}{11}g(x) - \frac{25}{11}\left(\frac{1}{10}x + \frac{7}{25}\right)\underbrace{\left(10x^2 + 12x + 2\right)}_{f(x)-(x-4)g(x)}$$

Substituting the last term

$$x + 1 = \left(\frac{5x^2}{22} - \frac{3x}{11} - \frac{3}{11}\right)g(x) + \left(\frac{-5x}{11} - \frac{7}{11}\right)f(x)$$

Thus we found over $\mathbb{Q}[x]$

For $\mathbb{Z}_2[x]$ we have to rewrite $f(x)$ and $g(x)$ and we get

$$f(x) = x^4 + x$$

$$g(x) = x^3 + 1$$

Notice that $x^4 + x = x(x^3 + 1)$. Meaning $g(x) \mid f(x)$ and thus $\gcd(f(x), g(x)) = g(x) = x^3 + 1$

For $\mathbb{Z}_{11}[x]$ we can borrow from a previous problem and see

$$f(x) = (x-4)g(x) + (10x^2 + 12x + 2)$$

Rewriting for $\mathbb{Z}_{11}$

$$f(x) = (x-4)g(x) + (-x^2 + x + 2)$$

Performing the algorithm again

$$g(x) = (-x^2 + x + 2)(-x - 5)$$

Hence we found our $\gcd(f(x), g(x)) = (-x^2 + x + 2)$. We can write

$$-x^2 + x + 2 = f(x) - (x-4)g(x)$$

6 Constructing a field of order 4

Recall that if k is a field, then $p(x) \in k[x]$ is irreducible if and only if $k[x]/\langle p(x)\rangle$.

Consider the polynomial ring, $\mathbb{Z}_2[x]$. Note that x^2+x+1 is irreducible. Because if it factored, it would factor into two, linear roots. But the only roots that it could possibly be in $\mathbb{Z}_2$ are $x=0$ and $x=1$. Meaning $\mathbb{Z}_2/\langle x^2+x+1\rangle$ is a field. We list all the elements and we find that it's just

$$\mathbb{Z}_2/\langle x^2+x+1\rangle = \{0, 1, \alpha, \alpha+1\}$$

And thus we can write it as

$$\mathbb{Z}_2(\alpha) = \{a+b\alpha \mid a, b \in \mathbb{Z}_2, \alpha^2 = \alpha+1\}$$

We then fill in the multiplication tables as such

$+$	0	1	α	$\alpha+1$
0	0	1	α	$\alpha+1$
1	1	0	$\alpha+1$	α
α	α	$\alpha+1$	0	1
$\alpha+1$	$\alpha+1$	α	1	0

$\cdot$	0	1	α	$\alpha+1$
0	0	0	0	0
1	0	1	α	$\alpha+1$
α	0	α	$\alpha+1$	1
$\alpha+1$	0	$\alpha+1$	1	α

A few facts we need to be aware of.

1. All finite field are of order p^r for some prime p
2. There is only 1 finite field up to isomorphism of any given order (all fields of the same order are isomorphic called $\mathbb{F}_{p^r}$)
3. To construct $\mathbb{F}_{p^r}$: Find an irreducible degree r polynomial $f(x) \in \mathbb{Z}_p[x]$, then

$$\mathbb{F}_{p^r} \cong \mathbb{Z}_p[x]/f(x)$$

7 $k[x]$ is a PID

Recall that the division algorithm tells us that for $f(x), g(x) \in k[x]$ then there exists a unique $q(x), r(x)$ such that

$$f(x) = g(x)q(x) + r(x)$$

Where $0 \leq \deg(r(x)) < \deg(g(x))$. We want to show that every ideal in $k[x]$ is a principal ideal and thus $k[x]$ is a PID.

Suppose $I \subseteq k[x]$ is an ideal. Take $p(x) \in I$ where $p(x)$ is monic and the degree $p(x)$ is minimal over all polynomials of positive degree (this $\langle p(x) \rangle \subseteq I$). Suppose $f(x) \in I$. Perform the division algorithm with $f(x)$ and $p(x)$ and we see

$$f(x) = p(x)q(x) + r(x)$$

Where $0 \leq \deg(r(x)) < \deg(p(x))$. Meaning $\deg(r(x)) = 0$ to preserve minimality of $\deg p(x)$. Suppose $r(x) = 0$. Then $f(x) \in \langle p(x) \rangle$ and thus $I \subseteq \langle p(x) \rangle$. If $r(x) = \alpha \neq 0 \in k$. Meaning α is a unit. Hence $\langle p(x) \rangle = \langle \alpha \rangle = k[x]$. Hence $I \in k[x]$.

Consider the ring $\mathbb{Z}[x]$. It is **not** a PID. Consider the ideal

$$I = \langle x, 2 \rangle = \{a_n x^n + ... + a_1 x + 2a_0 \mid a_i \in \mathbb{Z}\}$$

Suppose $p(x) \in \mathbb{Z}[x]$ where $\langle p(x) \rangle = I$. Thus

$$2 \in \langle p(x) \rangle \implies 2 = p(x)f(x) \implies \deg p(x) = \deg f(x) = 0$$

Hence $p(x) = 1$ or $p(x) = 2$. If $p(x) = 1$ then $I = \mathbb{Z}[x]$, a contradiction. Thus $p(x) = 2$ meaning $I = \langle 2 \rangle$ and hence $x \notin I$. But this contradicts x being a generator of I. Hence we found an ideal in $\mathbb{Z}[x]$ that is not principal.

And that's a good place to stop